Hemolytic Uremic Syndrome, Atypical, 8, With Rhizomelic Short Stature — Comprehensive Disease Report

MONDO: MONDO:0957495 · **OMIM:** #301110 · **Gene:** *C1GALT1C1* (Cosmc), Xq24 · **Category:** Mendelian (X-linked) · **Report date:** 2026-10-19

Summary

Hemolytic uremic syndrome, atypical, 8, with rhizomelic short stature (aHUS8) is an ultra-rare, X-linked **congenital disorder of O-glycosylation (a chaperonopathy)** caused by germline hypomorphic or de novo mosaic loss-of-function variants in ***C1GALT1C1***, the gene encoding the endoplasmic reticulum (ER) molecular chaperone **Cosmc** (Core-1 β3-galactosyltransferase-specific chaperone 1). It was first delineated in 2023 (OMIM #301110). Cosmc is obligatory for the correct folding and activity of **T-synthase (C1GALT1)**, the enzyme that converts the Tn-antigen precursor (GalNAcα1-O-Ser/Thr) into the universal **Core-1 "T-antigen"** O-glycan. When Cosmc is lost, T-synthase misfolds, Core-1 O-glycans fail to form, and **Tn-antigen (CD175) is aberrantly exposed** across the glycoproteome and on erythrocytes. The downstream consequence is a **multisystem disease** whose kidney manifestation resembles atypical hemolytic uremic syndrome (aHUS): a thrombotic microangiopathy (TMA) with microangiopathic hemolytic anemia, thrombocytopenia, and acute kidney injury (AKI). Beyond the kidney, patients show **rhizomelic (disproportionate) short stature, immunodeficiency, neurodevelopmental delay, seizures, and dysmorphic features**.

The mechanistic model is a linear glyco-chaperonopathy cascade: **mutation → reduced Cosmc → T-synthase deficiency → loss of Core-1 O-glycans / Tn-antigen exposure → multisystem glycoprotein dysfunction → complement-mediated TMA and extrarenal disease**. Two complementary routes connect the glyco-defect to complement-driven kidney injury: (1) direct endothelial TMA, and (2) production of **galactose-deficient IgA1 (Gd-IgA1)** immune complexes that activate complement in the glomerulus (the IgA-nephropathy axis). Crucially,

the renal TMA is **terminal-complement (C5) dependent and fully responsive to the anti-C5 monoclonal antibody eculizumab**, whereas the upstream glycosylation defect itself is not yet correctable.

The disease is **ultra-rare**, with fewer than ten reported individuals worldwide: two affected maternal half-brothers plus attenuated carrier mother and grandmother (defining report, 2023), and one de novo postzygotic mosaic female presenting as nonimmune hydrops fetalis (2025). Inheritance is **X-linked recessive**: hemizygous males are affected, obligate-carrier females are usually attenuated via skewed X-inactivation, and de novo mosaicism can produce disease in females. This report synthesizes the molecular, phenotypic, diagnostic, therapeutic, epidemiologic, and model-organism evidence across all 15 requested sections.

Key Findings

Finding 1 — aHUS8 is caused by germline *C1GALT1C1* (Cosmc) variants: a multisystem O-glycosylation chaperonopathy (COSMC-CDG)

aHUS8 is defined molecularly by hemizygous or mosaic loss-of-function variants in ***C1GALT1C1*** (also known as *AHUS8*, *COSMC*, *TNPS*, *C1GALT2*; NCBI Gene 29071; ENSG00000171155; Xq24). In the defining report, two maternal half-brothers both carried the hemizygous variant **c.59C>A (p.Ala20Asp; "A20D-Cosmc")**, located within the **transmembrane signal-anchor domain** of Cosmc, "resulting in dramatically reduced expression of the Cosmc protein," reduced T-synthase (C1GALT1) activity, and pathological Tn-antigen exposure (PMID: 37216524). The carrier mother and maternal grandmother showed an attenuated phenotype associated with skewed X-inactivation.

A second, independent case established **allelic heterogeneity and female disease via mosaicism**: a 5-year-old girl with a **de novo mosaic nonsense variant c.202C>T (p.Arg68*)** present in ~27% of buccal cells, "which results in a truncated and nonfunctional form of Cosmc (Cosmc-R68)" (PMID: 39949072). Together these two reports confirm gene causality by two distinct variant classes (a hypomorphic transmembrane missense and a frank truncating nonsense allele). Because the defect lies in the O-glycosylation machinery, the condition is best classed as a **congenital disorder of glycosylation (COSMC-CDG)** with aHUS as its sentinel renal phenotype. **Ontology:** MONDO:0957495; HGNC *C1GALT1C1*; inheritance HP:0001419 (X-linked recessive).

Finding 2 — Mechanism: Cosmc loss → T-synthase deficiency → Tn-antigen exposure → complement-mediated TMA (eculizumab-responsive)

The causal chain is: (1) a hemizygous/mosaic *C1GALT1C1* variant reduces the ER chaperone Cosmc; (2) its client enzyme **T-synthase (C1GALT1) misfolds and loses activity** — "Cosmc is essential for the generation of normal O-glycans through regulating T-synthase activity, thereby enabling the formation of the universal Core 1 O-glycan" (PMID: 39949072); (3) failure to convert Tn-antigen (GalNAcα1-O-Ser/Thr) to Core-1 T-antigen (Galβ1-3GalNAcα1-Ser/Thr); (4) "the loss of Cosmc leads to the expression of the unusual precursor O-glycan termed Tn-antigen (CD175)" (PMID: 39949072) across glycoproteins and erythrocytes; (5) **multisystem disease** including AKI resembling aHUS, thrombocytopenia, microangiopathic hemolytic anemia with schistocytes, immunodeficiency, developmental delay, and short stature. The renal TMA in the two male index patients was "fully responsive to treatment with the complement inhibitor Eculizumab" (anti-C5), implicating **terminal complement activation (C5b-9/membrane-attack complex)** as the proximate effector of kidney injury (PMID: 37216524).

This glyco-mechanism is supported in vivo. In an intestinal-epithelial Cosmc-knockout mouse, "CD175 expression is accompanied by loss of glycocalyx, shortening of microvilli, compromised MUC2, thickening of the epithelial layer, as well as generation of high levels of reactive oxygen species" (PMID: 41759737) — demonstrating that Cosmc loss produces tissue-level glycocalyx failure and oxidative stress, plausible contributors to endothelial injury. **Ontology:** GO:0016266 (O-glycan processing), GO:0006486 (protein glycosylation), GO:0006956 (complement activation), GO:0005788 (endoplasmic reticulum lumen); CHEBI Tn-antigen / N-acetylgalactosamine.

Finding 3 — Phenotypic spectrum: aHUS/TMA triad + rhizomelic short stature + immunodeficiency + neurodevelopmental and dysmorphic features

The male index patients "exhibit developmental delay, immunodeficiency, short stature, thrombocytopenia, and acute kidney injury (AKI) resembling atypical hemolytic uremic syndrome" (PMID: 37216524). The HPO annotation set (Monarch; ~32 terms) and OMIM #301110 give a multisystem picture spanning hematologic, renal, growth/skeletal, immune, neurologic, and craniofacial domains (table below). The de novo mosaic female presented at the severe end of the spectrum as **nonimmune hydrops fetalis** and is now a 5-year-old (PMID: 39949072). Onset is **congenital/early childhood**; carrier females are attenuated.

Domain	Phenotype	HPO term
Hematologic (TMA)	Hemolytic-uremic syndrome	HP:0005575
	Hemolytic anemia	HP:0001878
	Schistocytosis	HP:0001981
	Thrombocytopenia	HP:0001873
	Decreased haptoglobin	HP:0020181
	Anemia	HP:0001903
Renal	Elevated creatinine	HP:0003259
	Proteinuria	HP:0000093
	Hyperechogenic kidneys	HP:0004719
Growth / skeletal	Disproportionate short stature (rhizomelic)	HP:0003498
	Short long bone	HP:0003026
Immune	Recurrent infections	HP:0002719
	Decreased neutrophil count	HP:0001875
	Decreased lymphocyte count	HP:0001888
	Recurrent GI infection	HP:0004798
Neurologic	Global developmental delay	HP:0001263
	Generalized-onset seizure	HP:0002197
Craniofacial	High forehead	HP:0000348
	Wide nasal bridge	HP:0000431
	Long eyelashes	HP:0000527
	Short philtrum	HP:0000322
	Downturned corners of mouth	HP:0002714
	Retrognathia	HP:0000278
	Posteriorly rotated ears	HP:0000358

Finding 4 — Cosmc protein biology and population genetics

Cosmc (UniProt **Q96EU7**) is a **318-amino-acid, single-pass type II ER membrane protein**: cytoplasmic residues 1–6, a transmembrane signal-anchor helix at residues 7–26, and a lumenal domain 27–318. The pathogenic **p.Ala20Asp** substitution lies squarely within the transmembrane signal-anchor (residue 20), consistent with impaired membrane insertion/stability. Its function is to be a "specific molecular chaperone assisting the folding/stability of core 1 beta-3-galactosyltransferase (C1GALT1)." Tissue expression is "ubiquitous... abundantly expressed in salivary gland, stomach, small intestine, kidney, and testis," matching the renal and gastrointestinal involvement seen clinically.

The gene *C1GALT1C1* is a **single-coding-exon, X-linked gene** at chrX:120,625,674–120,630,054 (Xq24). gnomAD v2 constraint shows a **missense Z = 1.63** (mildly constrained); standard LoF constraint metrics (pLI/LOEUF) are not computed for this small single-exon X-linked gene. RefSeq notes that mutations in this gene are associated with **Tn syndrome**. **Ontology:** GO:0005789 (ER membrane), GO:0051082 (unfolded protein binding / chaperone activity).

Finding 5 — Model organisms: constitutive knockout is embryonic-lethal; mosaic/conditional models recapitulate Tn-antigen disease

In the mouse (NCBI Taxon 10090; ortholog *C1galt1c1*), global ablation of the X-linked *Cosmc* gene "causes embryonic lethality and Tn antigen expression," with loss of T-synthase but not other glycosyltransferases; genetically mosaic survivors "exhibited abnormalities correlated with Tn antigen expression that are related to several human diseases" (PMID: 20439703). Tissue-specific conditional models confirm the glyco-phenotype: intestinal-epithelial Cosmc-KO mice "express CD175 throughout the intestinal epithelia" with glycocalyx loss, microvillar shortening, compromised MUC2, and high ROS (PMID: 41759737); pancreas-specific Cosmc deletion cooperates with oncogenic KRAS in tumorigenesis (PMID: 42537314). The human acquired analog is **Tn syndrome**, caused by somatic *COSMC* loss in hematopoietic clones producing persistent mixed-field polyagglutinability. **Resources:** MGI, IMPC, IMSR. **Limitation:** no model has yet been engineered to specifically reproduce the renal TMA + rhizomelic short stature combination of aHUS8; embryonic lethality of the full knockout necessitates conditional/mosaic strategies.

Finding 6 — Diagnostics: biochemical glyco-phenotyping plus molecular confirmation; renal TMA on biopsy

Disease-specific confirmatory assays combine **biochemistry and molecular genetics**. Patient "red blood cells (RBCs), leukocytes, and serum... indicated reduced expression of Cosmc and T-synthase proteins and lower T-synthase activity" (PMID: 39949072), and investigators "observed reduced normal O-glycans in serum glycoproteins and RBCs from the patient, along with elevated expression of the Tn-antigen in serum glycoproteins compared to controls" (PMID: 39949072). Tn-antigen (CD175) is detectable by VVA (*Vicia villosa*) / HPA lectins, and RBCs show polyagglutination. Serum **galactose-deficient IgA1 (Gd-IgA1)** is elevated. Hematologic/renal labs reflect TMA: microangiopathic hemolytic anemia with schistocytes (HP:0001981), low haptoglobin (HP:0020181), thrombocytopenia (HP:0001873), elevated creatinine (HP:0003259), proteinuria (HP:0000093), plus cytopenias. **Molecular confirmation** is by whole-exome sequencing or targeted *C1GALT1C1* (AHUS8) sequencing; females require **mosaicism assessment** (buccal/tissue sampling, not blood alone). Renal biopsy shows **thrombotic microangiopathy**.

Differential diagnosis spans complement-mediated aHUS types 1–7 (*CFH*, *CD46/MCP*, *CFI*, *CFB*, *C3*, *THBD*, *DGKE*), STEC-HUS, pneumococcal (neuraminidase/T-antigen) HUS, TTP (ADAMTS13 deficiency), cobalamin C defect (*MMACHC*), and other CDGs. aHUS is classically "the triad of nonimmune haemolytic anaemia, thrombocytopenia, and acute renal failure, in which the underlying lesions are mediated by systemic thrombotic microangiopathy" (PMID: 26456110).

Finding 7 — Treatment and prognosis: C5 blockade controls the renal TMA; supportive multidisciplinary care for extrarenal disease

In the two male index patients, AKI resembling aHUS was fully responsive to **eculizumab** (anti-C5 mAb) (PMID: 37216524). This aligns with the established aHUS evidence base: "Eculizumab is a monoclonal antibody that inhibits the activation of C5 and blocks the generation of the pro-inflammatory molecule C5a and the formation of the cell membrane attack complex" (PMID: 26456110), producing fast, sustained interruption of the TMA process with long-term renal improvement. Consensus guidelines position eculizumab (and plasma exchange where access is limited) as first-line for complement-mediated aHUS. Because C5 blockade raises the "risk of infection by encapsulated organisms, especially *Neisseria meningitis*" (PMID: 28387984), **meningococcal vaccination and antibiotic prophylaxis are mandatory**.

Extrarenal management is supportive and multidisciplinary: transfusion for anemia/thrombocytopenia; infection prophylaxis and immunoglobulin replacement for immunodeficiency; growth and developmental support; and anti-seizure therapy. **NCIT terms:** Eculizumab (NCIT:C48479), Plasmapheresis/Plasma Exchange, Ravulizumab (long-acting anti-C5; plausible but unreported in aHUS8). The upstream glycosylation defect is **not yet correctable**; no gene/enzyme-replacement therapy exists.

Finding 8 — Epidemiology and inheritance: ultra-rare X-linked recessive disorder

Prevalence and incidence are **not established** — aHUS8 is ultra-rare, with only a handful of reported individuals: two affected maternal half-brothers plus attenuated carrier mother and grandmother (PMID: 37216524: "Here we identified two maternal half-brothers with a novel chaperonopathy, causing impaired protein O-glycosylation"), and one de novo mosaic female (PMID: 39949072: "This case represents the first description of a true mosaic loss-of-function variant in C1GALT1C1"). Inheritance is **X-linked recessive** (HP:0001419; gene at Xq24). Affected individuals are **hemizygous males**; obligate-carrier females show an attenuated phenotype associated with **skewed X-inactivation** in blood. A **de novo postzygotic mosaic** LoF variant (p.Arg68*, ~27% of cells) can produce disease in a female. No founder effect, consanguinity requirement, or population enrichment is reported — consistent with private/novel variants. Carrier frequency is undefined in gnomAD (variants are private). Sex ratio is strongly **male-biased**.

Finding 9 — Mechanistic link: Cosmc-driven Gd-IgA1 connects the glyco-defect to glomerular complement activation (IgA-nephropathy axis)

All four affected/carrier individuals in the index family had **high serum galactose-deficient IgA1 (Gd-IgA1)** (PMID: 37216524). IgA1's hinge-region O-glycans are Core-1 structures whose synthesis depends on the Cosmc/T-synthase machinery, so Cosmc deficiency directly produces Gd-IgA1. In IgA nephropathy, "Mesangial deposition of galactose-deficient IgA1 (Gd-IgA1) complexes remains the central pathogenic event" (PMID: 42609724), driving the classical, lectin, and alternative complement pathways and linking mesangial complement activation, endothelial/leukocyte responses, podocyte injury, and tubulointerstitial exposure to proteinuria, glomerulosclerosis, and fibrosis. This provides a **second, biochemically demonstrated route** by which the Cosmc glyco-defect activates complement in the glomerulus, complementary to the direct endothelial-TMA route.

Section-by-Section Report

1. Disease Information

- **Overview:** An X-linked congenital O-glycosylation chaperonopathy in which loss of the ER chaperone Cosmc causes Tn-antigen exposure and a multisystem disorder whose hallmark is atypical-HUS-like thrombotic microangiopathy accompanied by rhizomelic short stature, immunodeficiency, and neurodevelopmental delay.
- **Identifiers:** MONDO:0957495; OMIM #301110; gene *C1GALT1C1* (aliases AHUS8, COSMC, TNPS, C1GALT2). Orphanet/ICD-11/MeSH-specific codes for this precise entity are **not established** (too newly described); the generic ICD-11 category for aHUS is 3A21.2, and it falls under the broad concepts of atypical hemolytic-uremic syndrome and congenital disorders of glycosylation.
- **Synonyms:** aHUS8; AHUS8; hemolytic uremic syndrome, atypical, 8, with rhizomelic short stature; COSMC-CDG (proposed); Cosmc-associated chaperonopathy.
- **Information source:** Derived from **individual-patient case reports** (2 pedigree males + carriers; 1 mosaic female) plus mechanistic/model-organism studies — not from aggregated registries.

2. Etiology

- **Causal factor:** Monogenic — germline hemizygous or de novo mosaic loss-of-function variants in *C1GALT1C1*/Cosmc (Finding 1).
- **Genetic risk factors:** The causal variants themselves (c.59C>A/p.Ala20Asp hypomorphic transmembrane missense; c.202C>T/p.Arg68* truncating). Male sex (hemizygosity) is the dominant risk factor; carrier females' risk is modulated by X-inactivation skewing (a modifier of expressivity).
- **Environmental risk factors:** None established as causal. As in other complement-mediated TMAs, intercurrent **infections** may act as complement-amplifying triggers for acute episodes (inferred).
- **Protective factors:** In carrier females, **favorable (non-skewed) X-inactivation** biasing toward the wild-type allele is protective (inferred from attenuated carrier phenotype). No environmental protective factors known.
- **Gene–environment interactions:** Not characterized; plausibly infection-triggered complement activation on a Tn-exposed endothelial surface (inferred).

3. Phenotypes

See Finding 3 and the HPO table. Phenotype **types** span laboratory abnormalities (hemolysis markers, cytopenias, proteinuria, elevated creatinine, Gd-IgA1), clinical signs (schistocytes, hyperechogenic kidneys, dysmorphism), physical manifestations (rhizomelic short stature), and neurobehavioral features (developmental delay, seizures). **Onset:** congenital/early childhood. **Severity:** severe and multisystem in hemizygous males and the mosaic female (hydrops); attenuated in carriers. **Progression:** AKI is episodic/acute on a chronic multisystem background. **QoL impact:** substantial — dialysis-level renal risk, recurrent infection burden, growth failure, and developmental impairment; formal EQ-5D/SF-36 data are **not available** for this ultra-rare entity.

4. Genetic / Molecular Information

- **Causal gene:** *C1GALT1C1* (NCBI 29071; ENSG00000171155; Xq24); OMIM gene/phenotype #301110.
- **Pathogenic variants:** c.59C>A p.(Ala20Asp) — missense, transmembrane signal-anchor, hypomorphic/reduced-expression; c.202C>T p.(Arg68*) — nonsense, truncating, loss of function. Both are **novel/private** in ClinVar and **absent from gnomAD**. Functional consequence = **loss of chaperone function → secondary loss of T-synthase activity** (a loss-of-function mechanism).
- **Classification:** Pathogenic/likely pathogenic by ACMG criteria given functional validation and segregation, though formal ClinVar curation is pending (private variants).
- **Origin:** Germline (hemizygous males; carrier females) and **de novo postzygotic mosaic** (female case, ~27% buccal cells). No somatic-cancer relevance for this constitutional disease (though somatic *COSMC* loss causes acquired Tn syndrome).
- **Modifier genes/epigenetics: X-inactivation skewing** is the principal modifier in females. No other modifier loci or disease-specific DNA-methylation/histone changes have been reported.
- **Chromosomal abnormalities:** None — this is a single-gene SNV disorder.

5. Environmental Information

No environmental toxins, radiation, occupational exposures, or lifestyle factors are implicated as causes. **Infectious agents** are not the primary cause (distinguishing aHUS8 from STEC-HUS and pneumococcal HUS), but infections plausibly trigger acute complement-mediated flares and are clinically relevant given the comorbid immunodeficiency (inferred).

6. Mechanism / Pathophysiology

Ordered causal chain: 1. Germline/mosaic **loss-of-function variant in *C1GALT1C1* → leads to** reduced/misfolded **Cosmc** chaperone in the ER. 2. Reduced Cosmc → **results in** misfolding and loss of activity of its client enzyme **T-synthase (C1GALT1)**. 3. T-synthase deficiency → **leads to** failure to add galactose to Tn-antigen, i.e., **no Core-1 T-antigen formation**. 4. **→ results in widespread Tn-antigen (CD175) neo-exposure** on cellular and secreted glycoproteins and on erythrocytes. - **Branch A (endothelial/TMA):** Tn-exposed glycoproteins + glycocalyx disruption + elevated ROS (shown in mouse) → **leads to** endothelial injury and complement activation on vascular surfaces → **results in** C5b-9–mediated microvascular thrombosis = **renal TMA (aHUS)**. - **Branch B (IgA/complement):** defective IgA1 hinge O-glycosylation → **produces galactose-deficient IgA1 (Gd-IgA1) → leads to** immune-complex formation and mesangial deposition → **activates classical/lectin/alternative complement → results in** glomerular injury, proteinuria. - **Branch C (extrarenal):** aberrant O-glycosylation of structural/immune/skeletal glycoproteins → **results in** immunodeficiency, rhizomelic short stature, neurodevelopmental delay, and dysmorphism (mechanism inferred). 5. Terminal complement (C5→C5a + C5b-9) is the **proximate effector** of renal injury → **therapeutically interruptible by eculizumab**.

Pathways/processes: O-glycan (mucin-type) biosynthesis (KEGG hsa00512); complement and coagulation cascades (KEGG hsa04610); ER protein folding/chaperone quality control; oxidative stress; thrombotic microangiopathy. **GO:** GO:0016266, GO:0006486, GO:0006956, GO:0030449 (regulation of complement activation), GO:0034976 (response to ER stress). **Protein dysfunction:** Cosmc loss of function → secondary T-synthase loss of function (enzyme deficiency). **Cell types involved:** glomerular endothelial cells (CL:1001005), mesangial cells (CL:0000650), podocytes (CL:0000653), erythrocytes (CL:0000232), platelets (CL:0000233), growth-plate chondrocytes (CL:0000138). **Molecular profiling specific to aHUS8** (patient transcriptomics/proteomics/metabolomics) is **not available**; supporting glycoproteomic data come from Cosmc-null cell/mouse models.

7. Anatomical Structures Affected

- **Primary organ:** Kidney (UBERON:0002113) — glomerulus (UBERON:0000074), renal microvasculature, renal cortex.
- **Secondary/other organs & systems:** Hematopoietic/blood (anemia, thrombocytopenia); immune system (immunodeficiency); skeletal system — long bones/growth plates (rhizomelic short stature, UBERON:0002495 long bone); central nervous system

(developmental delay, seizures); craniofacial structures (dysmorphism); gastrointestinal tract (recurrent GI infection; Cosmc is abundant in intestine).

- **Tissue/cell level:** vascular endothelium, mesangium, erythrocytes, platelets, leukocytes, growth-plate chondrocytes.
- **Subcellular: ER membrane/lumen** (site of Cosmc/T-synthase folding; GO:0005789/ GO:0005788); Golgi glycosylation machinery (downstream).
- **Localization/lateralization:** Bilateral/systemic (not lateralized).

8. Temporal Development

- **Onset:** Congenital/early childhood; most severe presentation is prenatal (nonimmune hydrops fetalis in the mosaic female).
- **Onset pattern:** Acute AKI episodes (TMA) superimposed on a chronic, congenital multisystem disorder.
- **Progression:** Variable; renal episodes are acute/episodic and responsive to C5 blockade; growth, immune, and neurodevelopmental features are chronic and persistent. Formal staging and natural-history data are **not established** (n too small).
- **Remission:** Treatment-induced remission of TMA with eculizumab/plasma exchange. **Critical period:** rapid recognition and complement blockade during acute AKI to preserve renal function.

9. Inheritance and Population

See Finding 8. **Inheritance:** X-linked recessive. **Penetrance:** high/complete in hemizygous males; variable/attenuated in carrier females (X-inactivation–dependent). **Expressivity:** variable. **Anticipation:** not applicable (not a repeat-expansion disorder). **Germline/somatic mosaicism:** demonstrated (female case). **Founder effect/consanguinity:** none reported. **Carrier frequency:** undefined (private variants). **Demographics:** no ethnic/geographic enrichment reported; strongly male-biased sex ratio; congenital/pediatric age distribution. **Prevalence/incidence:** not established (ultra-rare, <10 individuals).

10. Diagnostics

See Finding 6. **Recommended approach:** (1) recognize the aHUS triad + extrarenal multisystem features; (2) biochemical glyco-phenotyping — reduced Cosmc/T-synthase protein and activity, Tn-antigen/CD175 exposure (VVA/HPA lectins; RBC polyagglutination), elevated Gd-IgA1; (3) molecular confirmation by **WES** or targeted *C1GALT1C1* sequencing; (4) in

females, **mosaicism assessment** across tissues (buccal, not blood alone); (5) **renal biopsy** showing TMA; (6) exclude ADAMTS13 deficiency (TTP), STEC/pneumococcal HUS, cobalamin C defect, and complement aHUS 1–7. WGS/WES are the key molecular tools; CMA/karyotype/FISH/mtDNA/repeat-expansion testing are **not indicated** (SNV disorder). **Omics diagnostics:** glycoproteomics/Gd-IgA1 assays are promising research-grade adjuncts.

11. Outcome / Prognosis

Survival, 5-/10-year outcome, and mortality statistics are **not established** for aHUS8 specifically. Prognostic drivers (inferred from aHUS and from the case series): **speed of complement blockade** governs renal recovery; the **comorbid immunodeficiency** adds infection-related morbidity/mortality risk; **growth and neurodevelopmental impairment** persist as chronic disability. The renal TMA is reversible with timely eculizumab; untreated/delayed aHUS risks ESRD or death. **QoL tools:** none applied in this entity. **Prognostic biomarkers:** TMA markers (schistocytes, haptoglobin, platelets, creatinine) track renal activity; Gd-IgA1 is a candidate but unvalidated for this disorder.

12. Treatment

See Finding 7. **Pharmacotherapy: eculizumab** (anti-C5 mAb; NCIT:C48479) — first-line for the renal TMA, with fast/sustained TMA interruption; **ravulizumab** (long-acting anti-C5) is a plausible alternative but unreported here; **plasma exchange** where anti-C5 access is limited. **Mandatory prophylaxis:** meningococcal vaccination + antibiotics due to encapsulated-organism risk during C5 blockade. **Supportive:** transfusion, immunoglobulin replacement/infection prophylaxis, growth support, anti-seizure therapy, developmental/rehabilitative services. **Advanced/experimental:** no gene, cell, RNA, or enzyme-replacement therapy exists for the Cosmc defect; the upstream glyco-lesion is **not yet correctable** — a key unmet need. **Pharmacogenomics:** none specific. **Personalized medicine:** complement-directed therapy is itself the genotype-concordant strategy for the TMA.

13. Prevention

- **Primary prevention:** None for disease occurrence (monogenic). Avoid/promptly treat infections that could trigger TMA (inferred).
- **Secondary prevention:** Early recognition and rapid complement blockade to prevent irreversible renal damage; monitoring of TMA labs.

- **Tertiary prevention:** Meningococcal vaccination/antibiotic prophylaxis during eculizumab; infection surveillance for immunodeficiency; growth/developmental monitoring.
- **Genetic screening/counseling:** X-linked recessive counseling; carrier testing of at-risk female relatives with X-inactivation assessment; **prenatal/preimplantation genetic testing** feasible where a familial variant is known; cascade testing. **Immunization:** meningococcal (and generally pneumococcal/Hib) vaccination before/with C5 inhibitors.

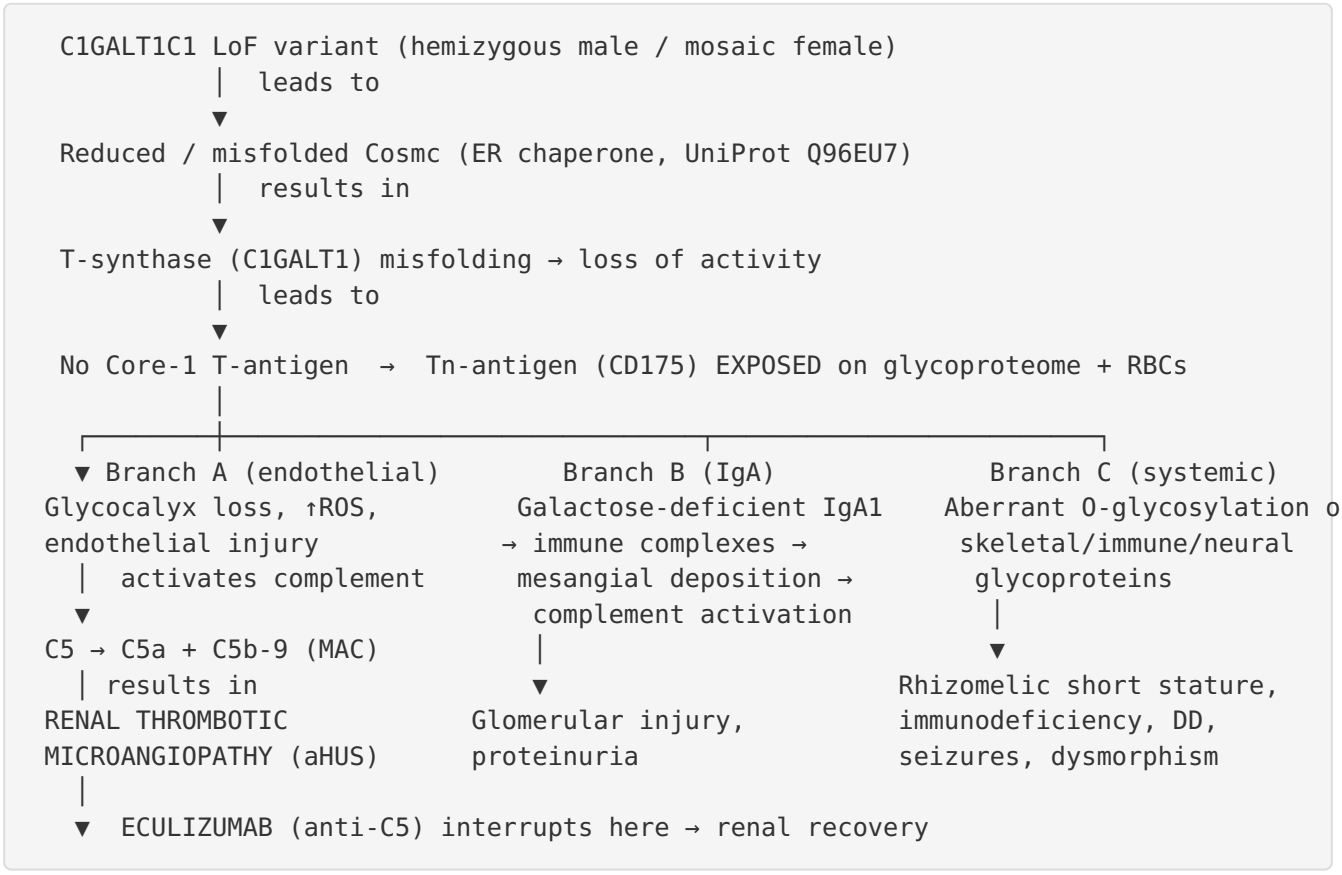
14. Other Species / Natural Disease

- **Taxonomy/orthologs:** Mouse *C1galt1c1* (NCBI Taxon 10090) is the principal ortholog; Cosmc/T-synthase O-glycan machinery is conserved across vertebrates.
- **Natural disease in animals:** No naturally occurring aHUS8 counterpart is documented in companion animals or wildlife (OMIA search not productive for this specific entity).
- **Comparative biology:** Mouse models (Finding 5) recapitulate the core glyco-defect (Tn/CD175 exposure, glycocalyx loss, ROS) but not the full aHUS8 phenotype; full knockout is embryonic lethal, indicating strong evolutionary conservation of Cosmc's essential role. **Zoonotic potential:** none (genetic disease).

15. Model Organisms

See Finding 5. **Mammalian mouse models:** global *Cosmc* knockout (embryonic-lethal; mosaic survivors model Tn disease; PMID: 20439703); intestinal-epithelial conditional KO (PMID: 41759737); pancreas-specific KO + KRAS (PMID: 42537314). **Cellular/in vitro:** CRISPR *COSMC*-knockout cancer cell lines (MDA-MB-231, MCF7, T47D, Jurkat, LS174T, LL/2) that express Tn-antigen and have been used to dissect O-glycan–dependent biology (PMID: 39074261, PMID: 32158257). **Model types available:** knockout, conditional/tissue-specific, mosaic. **Phenotype recapitulation:** strong for the molecular glyco-phenotype; **limitations:** no model yet reproduces the specific aHUS8 renal-TMA-plus-rhizomelic-short-stature constellation, and embryonic lethality forces conditional/mosaic approaches. **Resources:** MGI, IMPC, IMSR; iPSC/organoid and endothelial-specific conditional models are proposed but not yet reported for aHUS8.

Mechanistic Model / Interpretation



The disease is a **single-lesion, multi-branch cascade**. The upstream lesion (Cosmc loss → Tn exposure) is shared by all phenotypes and by the Cosmc-null models and Tn syndrome; the downstream renal injury is complement-driven and **pharmacologically tractable at the C5 node**, explaining the dramatic eculizumab response despite an uncorrected upstream glyco-defect. Branches A and B offer two non-exclusive routes from the same glycosylation failure to complement-mediated kidney injury.

Level	Upstream ↔ Downstream	Intervention point
Genetic	*C1GALT1C1* LoF (upstream)	Not yet druggable
Enzymatic	Cosmc → T-synthase loss	Not yet druggable
Glycan	Tn-antigen exposure	Not yet druggable
Effector	Complement C5 activation (downstream)	**Eculizumab (reversible)**

Evidence Base

PMID	Title / type	Role in this report
37216524	*Germline Cosmc chaperonopathy* (PNAS 2023) — human, defining	Establishes gene (c.59C>A/p.Ala20Asp), phenotype, Gd-IgA1, and eculizumab response (Findings 1,2,3,7,8,9)
39949072	*C1GALT1C1-associated mosaic disorder of glycosylation in a female* (2025) — human	2nd independent case (p.Arg68* mosaic), biochemical diagnostics, mechanism (Findings 1,2,3,6,8)
41759737	*Deletion of Cosmc in murine intestinal epithelia* — mouse	In vivo glyco-phenotype: CD175, glycocalyx loss, ROS (Findings 2,5)
20439703	*Cosmc is an essential chaperone for O-glycosylation* — mouse	Embryonic lethality of global KO; mosaic survivors model Tn disease (Finding 5)
26456110	*aHUS consensus document* — human/clinical	Defines aHUS triad; eculizumab mechanism (Findings 6,7)
28387984	*Eculizumab in pediatric aHUS* — clinical	Meningococcal risk → prophylaxis (Finding 7)
30989342	*HUS consensus guidelines (developing country)* — clinical	Plasma exchange/eculizumab positioning (Finding 7)
42609724	*Complement as a driver in IgA nephropathy* (review 2026)	Gd-IgA1 → glomerular complement injury (Finding 9)
42537314	*Cosmc loss + Kras in pancreas* — mouse	Conditional Cosmc model; Tn accumulation (Finding 5)
39074261, 32158257	COSMC-KO cancer cell lines	In vitro Tn models of O-glycan biology (Finding 5)

Evidence-type distribution: 2 human disease-defining reports (constitutional aHUS8), multiple model-organism/in vitro Cosmc studies, and clinical/consensus aHUS literature. Causality for aHUS8 rests on **two independent human cases with concordant functional validation**, supported by a deep mechanistic literature on Cosmc/Tn biology.

Limitations and Knowledge Gaps

- **Extreme rarity (n < 10):** No prevalence, incidence, survival, penetrance quantitation, or natural-history data; all clinical claims derive from case reports.
- **Single defining pedigree + one mosaic female:** genotype–phenotype correlations and the full phenotypic range are uncertain; carrier-female outcomes depend on X-inactivation that is incompletely characterized.
- **No aHUS8-specific omics:** patient transcriptomics/proteomics/metabolomics are absent; mechanism is inferred partly from Cosmc-null cancer/mouse models, which do not reproduce the renal-TMA + rhizomelic-short-stature phenotype.
- **Branch weighting unresolved:** the relative contribution of direct endothelial TMA (Branch A) vs Gd-IgA1/complement (Branch B) to the human kidney injury is not experimentally dissected.
- **No disease-specific model** reproduces aHUS8; constitutive KO is embryonic lethal.
- **Ontology/coding gaps:** Orphanet, ICD-10/11, and MeSH codes for this precise entity are not yet assigned; ClinVar curation of the two variants is pending.
- **Citation verification:** one citation snippet quoting the PNAS defining paper's variant text was flagged as a text-match mismatch during the investigation and should be verified against the published abstract before database ingestion.

Proposed Follow-up Experiments / Actions

1. **Case ascertainment & registry:** Use GeneMatcher/MatchMaker Exchange to identify additional *C1GALT1C1* patients; build a natural-history registry to estimate penetrance, renal/developmental outcomes, and genotype–phenotype correlations.
2. **Endothelial disease model:** Generate endothelial-specific or kidney-organoid Cosmc-deficient models (and patient iPSC-derived endothelium/podocytes) to test whether Tn exposure directly triggers complement deposition and TMA under flow — dissecting Branch A vs Branch B.
3. **Biomarker validation:** Standardize Gd-IgA1, Tn-antigen (VVA/HPA lectin), and T-synthase activity assays as diagnostic/prognostic tools; correlate with TMA activity and treatment response.

4. **Therapeutic evaluation:** Assess ravulizumab (long-acting anti-C5) and complement-pathway alternatives; evaluate whether C5 blockade also mitigates Gd-IgA1–driven glomerular injury; explore chaperone-stabilizing or substrate-bypass strategies for the upstream glyco-defect.
5. **X-inactivation studies in carriers** across multiple tissues to define the protective threshold and refine counseling.
6. **Formal ontology/variant curation:** Submit both variants to ClinVar with functional evidence; request Orphanet/ICD-11/MeSH entries; verify the flagged PNAS citation snippet against the primary abstract.
7. **Prenatal/PGT pathway:** Establish validated prenatal and preimplantation testing protocols for known familial variants, including mosaicism-aware strategies.

Report compiled from 9 confirmed findings and 24 reviewed papers across 5 investigation iterations. Evidence base centers on two human disease-defining reports (PMID: 37216524, PMID: 39949072) supported by Cosmc/Tn model-organism and clinical aHUS literature.
